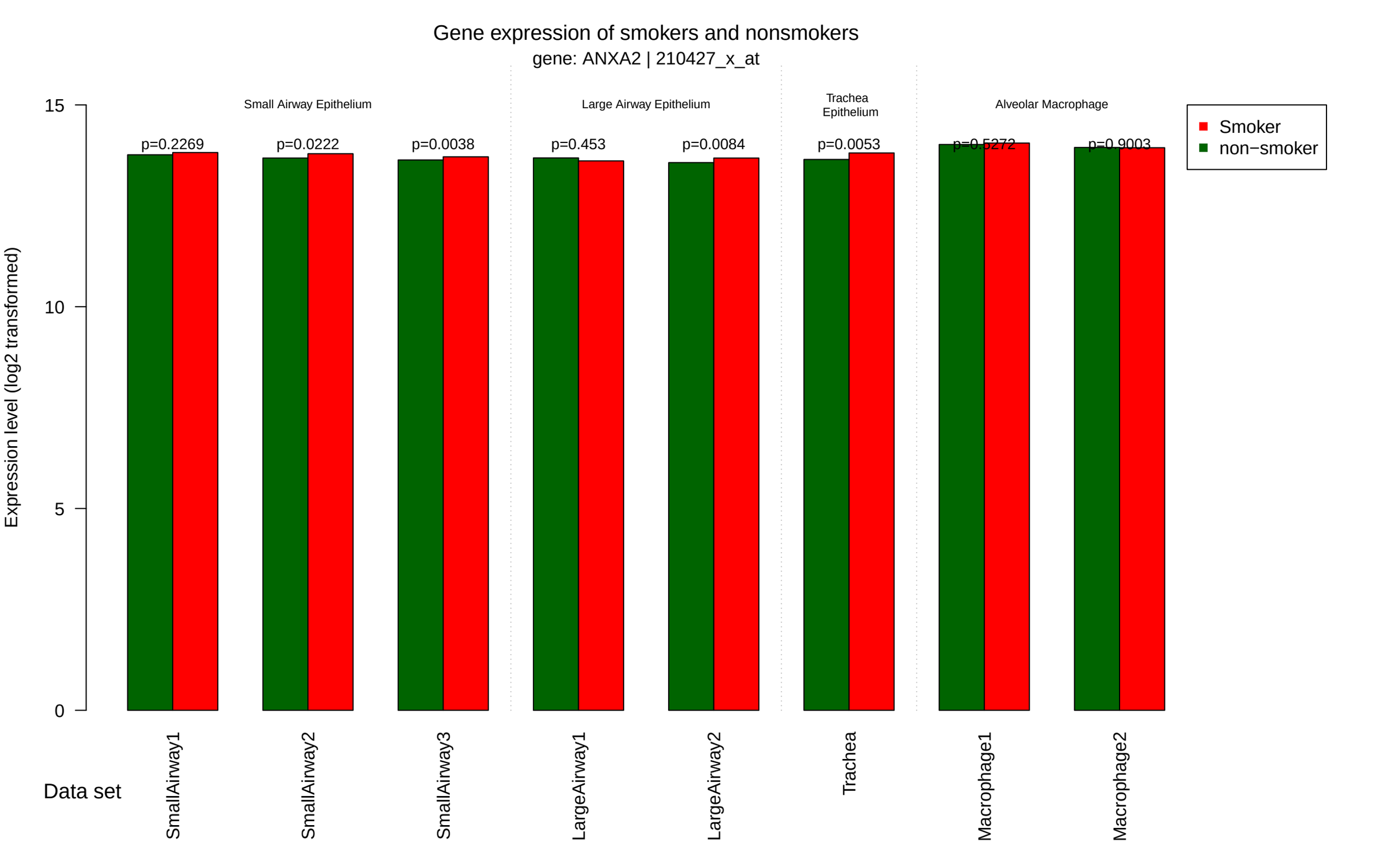
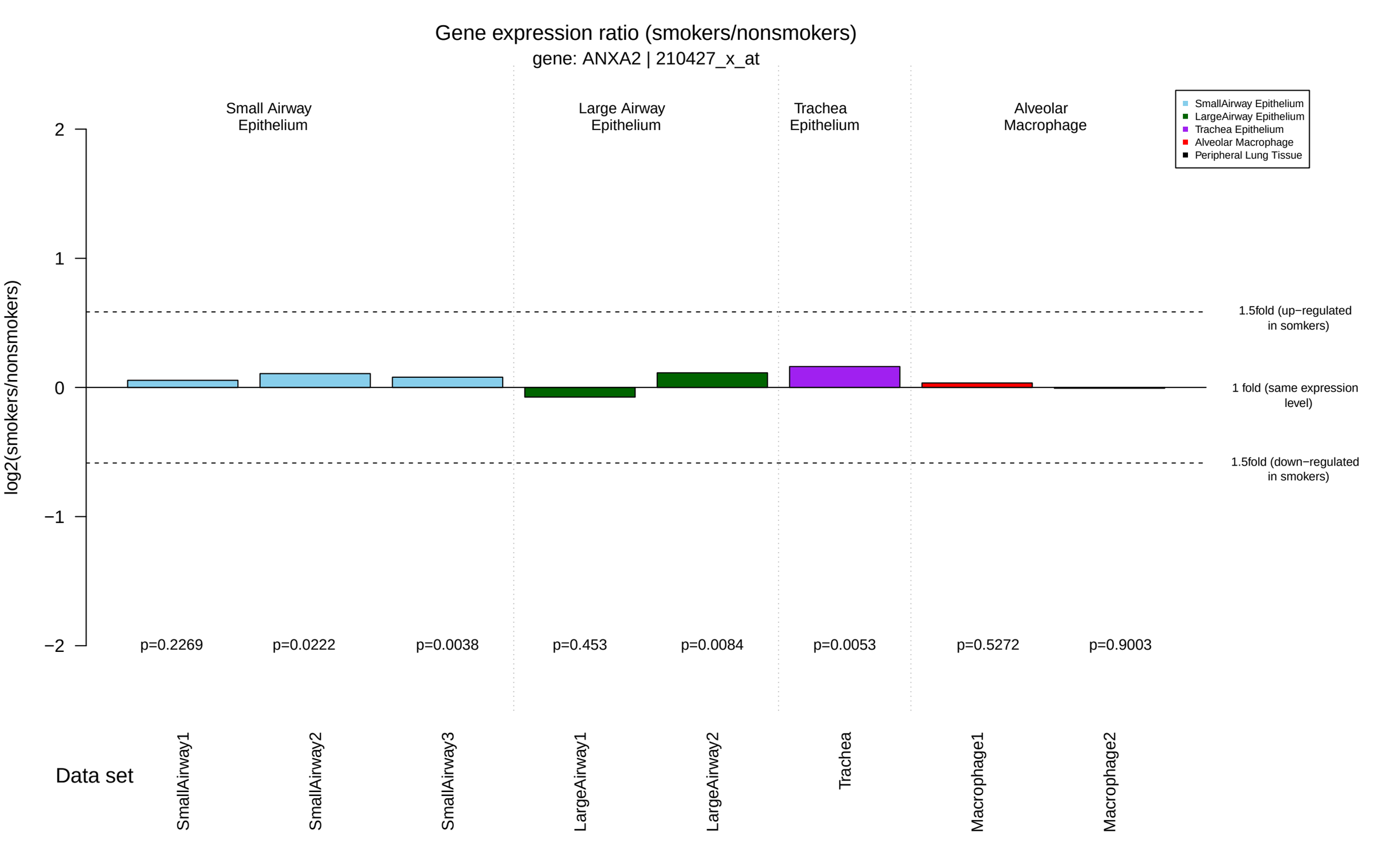
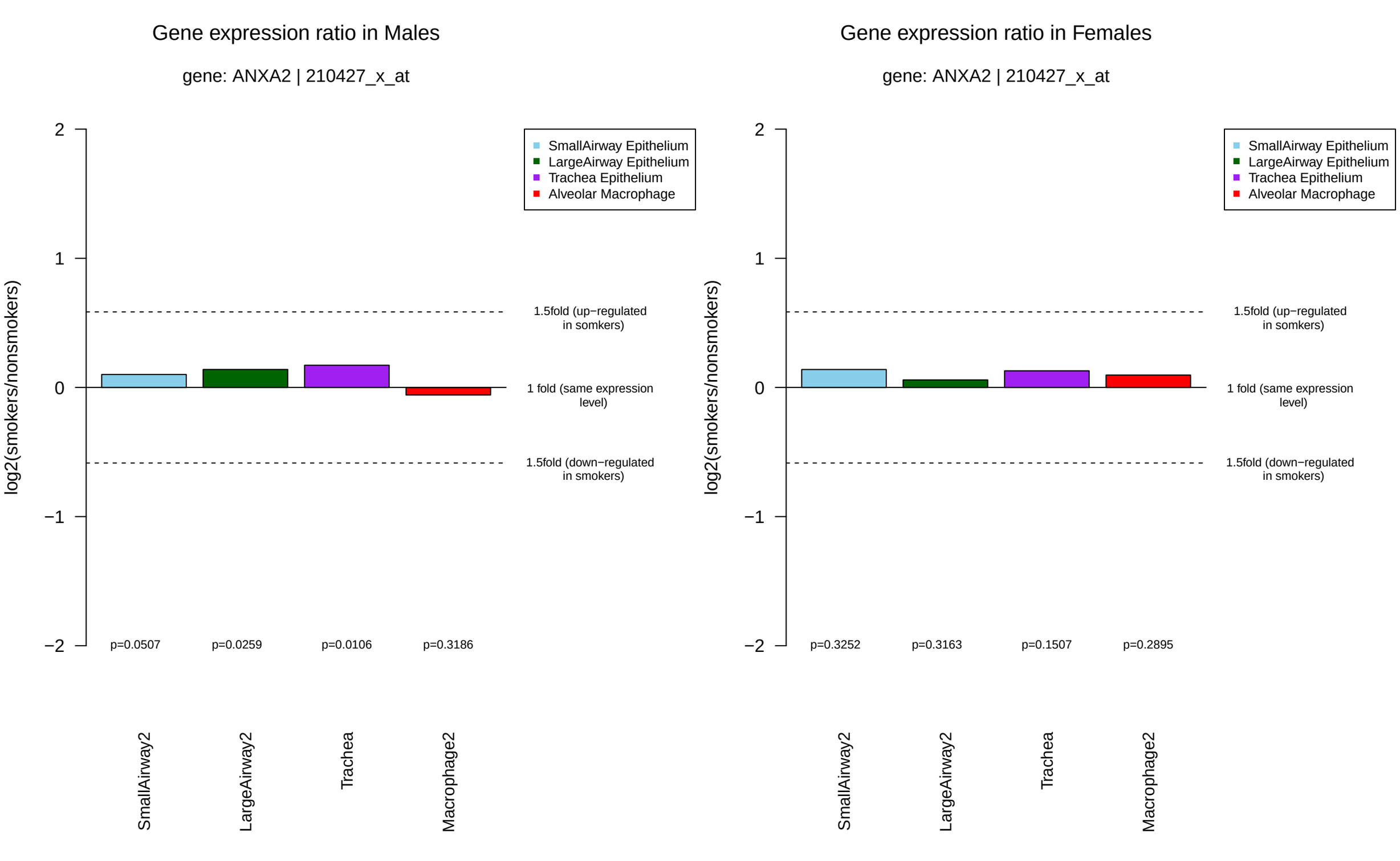
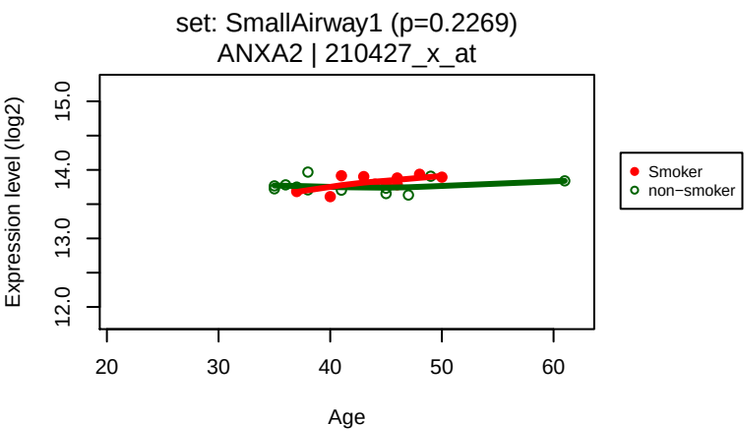
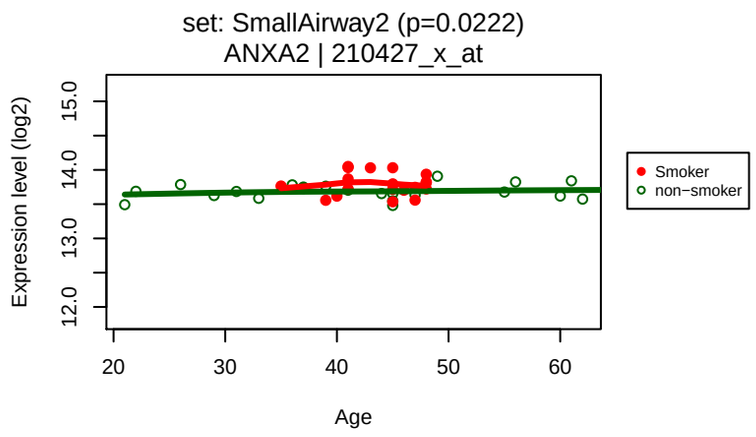
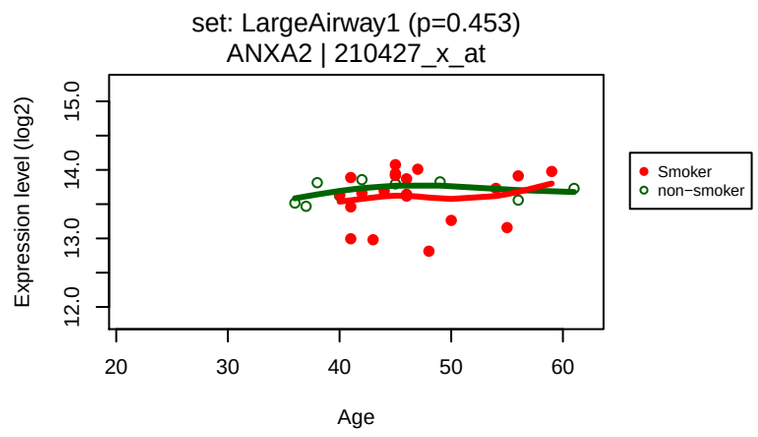

Gene expression of smokers and nonsmokers
gene: ANXA2 | 210427_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2269
p=0.0222
p=0.0038
p=0.453
p=0.0084
p=0.0053
p=0.5272
p=0.9003
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

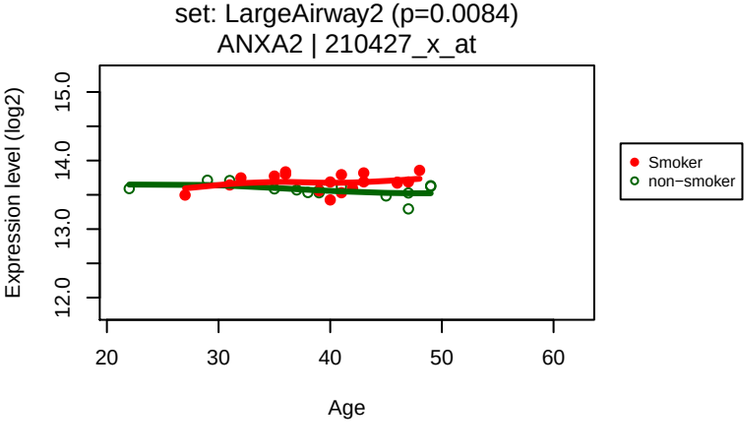

Gene expression ratio (smokers/nonsmokers)
gene: ANXA2 | 210427_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2269
p=0.0222
p=0.0038
p=0.453
p=0.0084
p=0.0053
p=0.5272
p=0.9003
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

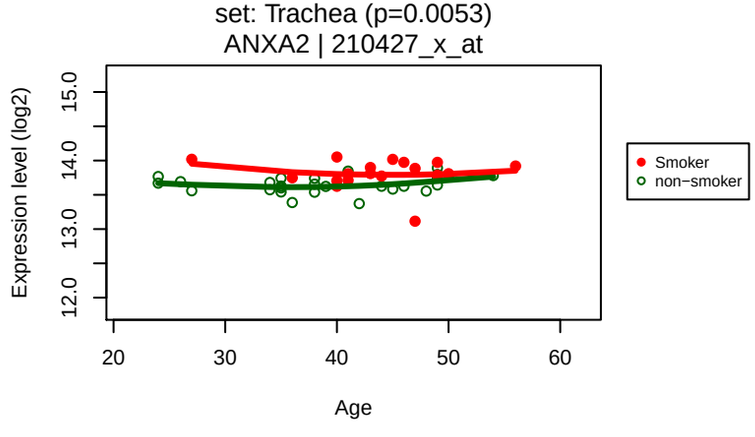

Gene expression ratio in Males
gene: ANXA2 | 210427_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0507
p=0.0259
p=0.0106
p=0.3186
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ANXA2 | 210427_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3252
p=0.3163
p=0.1507
p=0.2895
SmallAirway2
LargeAirway2
Trachea
Macrophage2

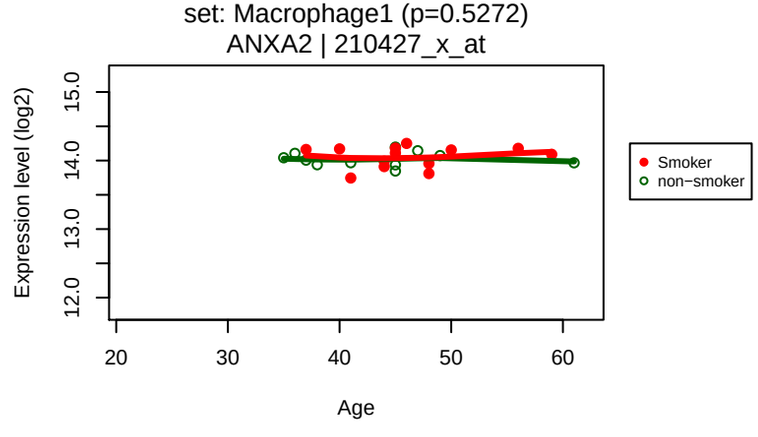

set: SmallAirway1 (p=0.2269)
ANXA2 | 210427_x_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker

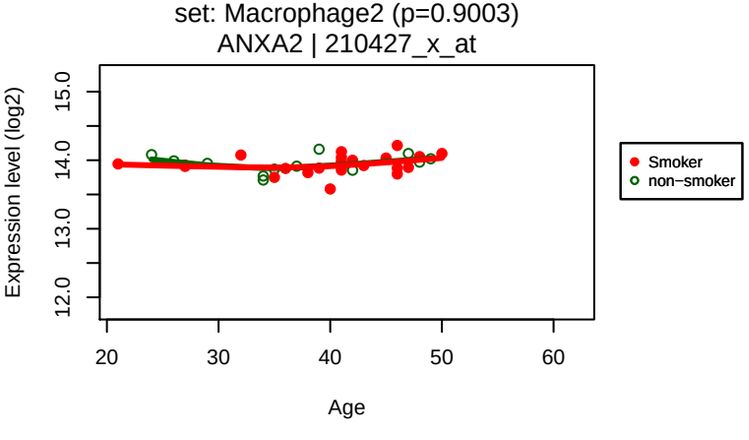

set: SmallAirway2 (p=0.0222)
ANXA2 | 210427_x_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker

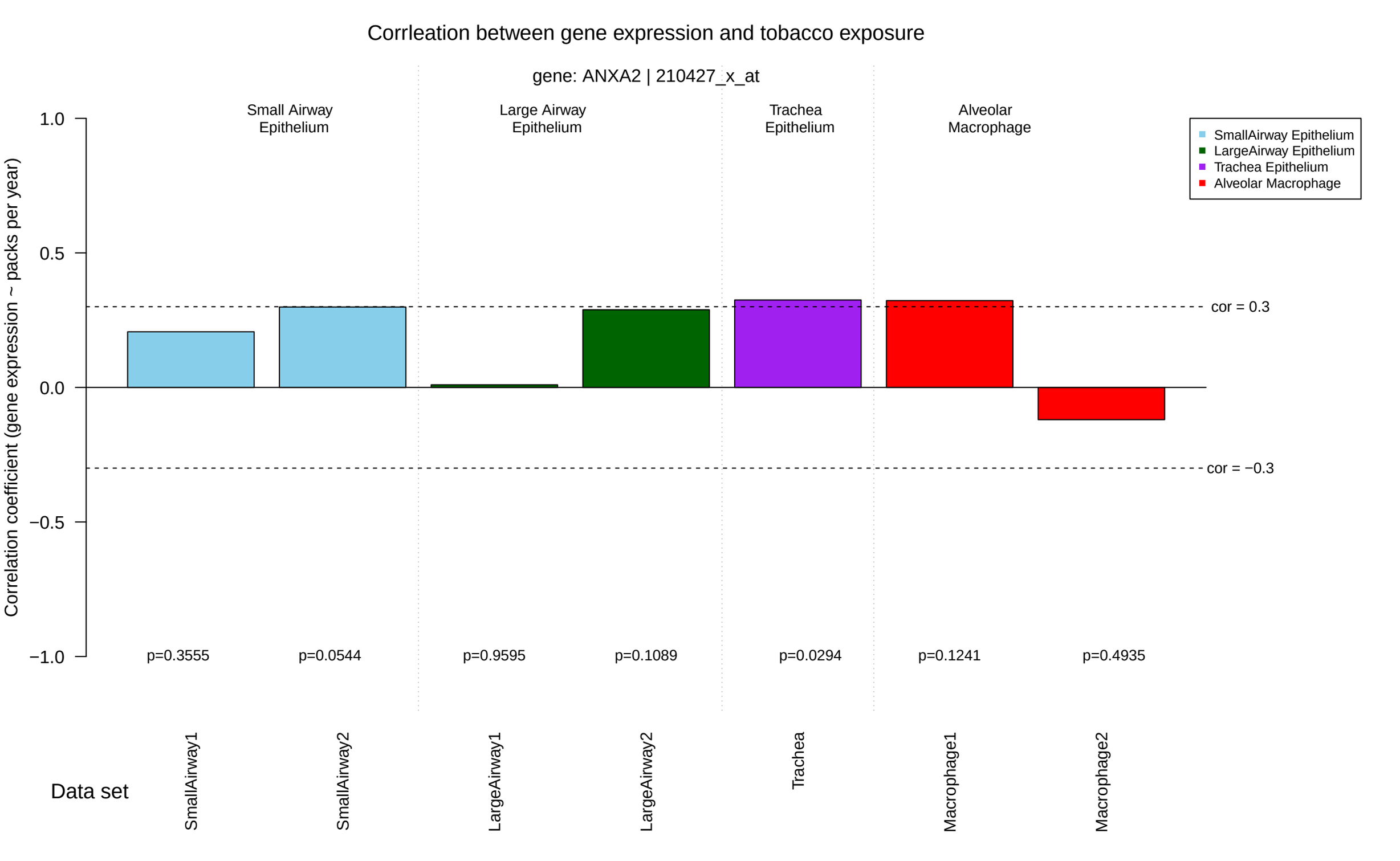

set: LargeAirway1 (p=0.453)
ANXA2 | 210427_x_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0084)
ANXA2 | 210427_x_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.0053)
ANXA2 | 210427_x_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5272)
ANXA2 | 210427_x_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9003)
ANXA2 | 210427_x_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ANXA2 | 210427_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3555
p=0.0544
p=0.9595
p=0.1089
p=0.0294
p=0.1241
p=0.4935
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2